



European Network of  
Transmission System Operators  
for Electricity

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# EDI BEST PRACTICES

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## Revision History

| Version | Release | Date       | Comments                                                                                                                                                                                                                                           |
|---------|---------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0       | 0       | 2017-01-27 | First drafting of the document based on maintenance request from WG EDI.                                                                                                                                                                           |
| 1       | 0       | 2017-02-23 | Add comments from WG EDI for Market Committee Approval                                                                                                                                                                                             |
| 1       | 1       | 2017-11-08 | Delete the recommendation on UUID                                                                                                                                                                                                                  |
| 2       | 0       | 2018-01-15 | Draft version of Market Committee approval for publication                                                                                                                                                                                         |
| 2       | 1       | 2019-09-10 | Added new chapter to advise about the mRID usage and warn users about the mRID enlargement from 35 to 60 characters.<br>Added a new chapter to explain how to properly use the Harmonised Electricity Market Role Model (HEMRM)<br>Approved by MC. |
| 2       | 2       | 2025-10-29 | Added a new chapter "Usage of revisionNumber"<br>ICTC approved                                                                                                                                                                                     |

36

## Objective

The purpose of this document is to provide the ENTSO-E EDI best practices document when using XML instances.

## XML instance recommendations

In order to ensure flexibility and independency from local implementation parameters, the following recommendations are to be applied:

- When providing an XML instance document to another party, it is recommended that the schema location instruction (and import of "http://www.w3.org/2001/XMLSchema-instance" namespace) is not used.
- To validate an XML document with the correct XML Schema, the document must declare its default namespace so that the appropriate schema can be identified and applied (see Figure 1 Example of XML instance with xmlns).

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <Acknowledgement_MarketDocument xmlns="urn:iec62325.351:tc57wg16:451-1:acknowledgementdocument:7:0">
3    <mRID>ACK-SOAM-SWI-RTE-091214-V1</mRID>
4    <createdDateTime>2014-12-10T09:38:30Z</createdDateTime>
5    <sender_MarketParticipant.mRID codingScheme="A01">10XFR-RTE-----Q</sender_MarketParticipant.mRID>
6    <sender_MarketParticipant.marketRole.type>A04</sender_MarketParticipant.marketRole.type>
7    <receiver_MarketParticipant.mRID codingScheme="A01">10XCH-SWISSGRIDC</receiver_MarketParticipant.mRID>
8    <receiver_MarketParticipant.marketRole.type>A04</receiver_MarketParticipant.marketRole.type>
9    <received_MarketDocument.mRID>20141209_SOAM_SGD_RTE</received_MarketDocument.mRID>
10   <received_MarketDocument.revisionNumber>1</received_MarketDocument.revisionNumber>
11   <received_MarketDocument.type>A46</received_MarketDocument.type>
12   <received_MarketDocument.createdDateTime>2014-12-10T09:38:30Z</received_MarketDocument.createdDateTime>
13   <Reason>
14     <code>A01</code>
15   </Reason>
16 </Acknowledgement_MarketDocument>

```

**Figure 1 Example of XML instance with xmlns**

- As all XML Schemas created with the European Style Market Profile methodology only use one namespace, there is no need to define a prefix for this namespace, but it should be accepted if included.

## Use of local codes in codelist

ENTSO-E provides a package with XML instance files defining the ENTSO-E codelists and an empty local extensions to these. This enables ones to define specific local codes to be used based on bilateral agreement with other parties, e.g. specific market rules for a local market.

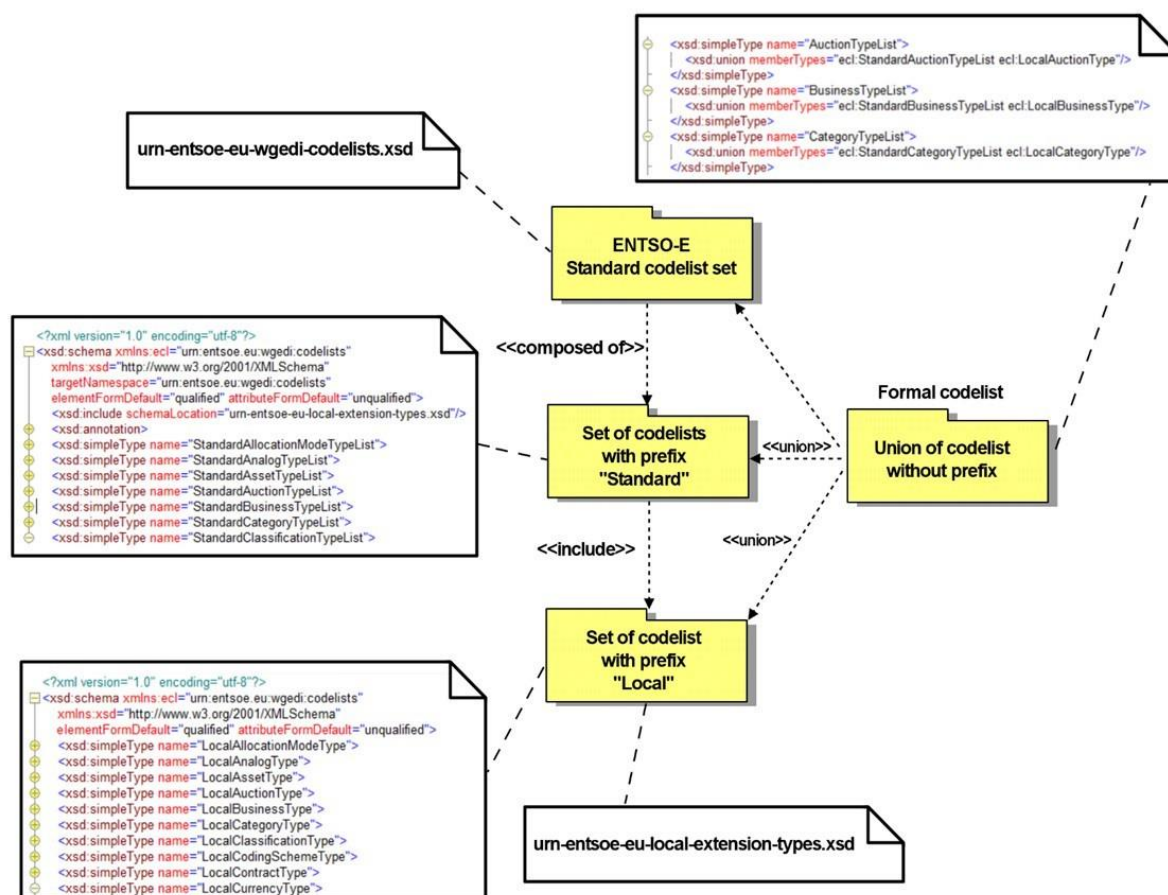
As can be seen in Figure 2 the standard codelist set is itself divided into two basic parts:

- The list of "standard" codelists that consists of all the codes that have been approved within CIM WG.
- A list of "local" codelists that are assigned by individual TSOs for internal local market use. These codes are local to a given market and are managed by the local TSO in order to satisfy local market constraints. These codes cannot be used outside the market in which they have been designed.

In order to provide a stable core component library, ENTSO-E makes available the library structure that includes an "empty" local set of codelists. This "empty" set of codelists in fact contains by default the first "standard" code that can be found within the "standard" codelist. This ensures that within the value space of the "local" codelist there is no possibility of validating codes that do not exist. Local codes should start with late letters in the alphabet. This local codelist schema has a standardised name within the ENTSO-E environment which is "urn-entsoe-eu-local-extension-types.xsd".

Each TSO then could replace the “empty” local set of codelists with the “local” codelists that are needed within its local marketplace. It is also the responsibility of the TSO to ensure the distribution of the “local” codelist to the market participants within the local market area.

The ENTSO-E XML schema environment “merges” the two codelist sets in order to provide at the electronic document level the complete set of codes possible (i.e. the union of the “standard” and “local” codes).



### Figure 2 – Codelist schema

## Time

The following rules are to be used concerning the time:

- All the dates and times within the MarketDocument shall be expressed in coordinated universal time (UTC) in compliance with ISO 8601 as YYYY-MM-DDThh:mm:ssZ.
- For all time intervals, the start date and time is included in the scope of the interval whereas the end date and time is excluded from the scope of the interval, i.e. [start date and time, end date and time[.
- For all time intervals, the start date and time as well as the end date and time are expressed in coordinated universal time (UTC) in compliance with ISO 8601 but without the seconds, i.e. as YYYY-MM-DDThh:mmZ.

## mRID usage advise

Please, check in which identification scheme is the mRID of the object.

If it is an EIC coding scheme, please refer to the EIC Scheme Terms and Conditions (<https://www.entsoe.eu/data/energy-identification-codes-eic/>) and check the existence

96 of the code on the CIO ([https://www.entsoe.eu/data/energy-identification-codes-eic/eic-](https://www.entsoe.eu/data/energy-identification-codes-eic/eic-approved-codes/)  
97 [approved-codes/](https://www.entsoe.eu/data/energy-identification-codes-eic/eic-approved-codes/)) or LIO repositories.

98 mRID of Document, Series, TimeSeries, Auction and MarketAgreement

99 mRID of Document, Series, TimeSeries, Auction and MarketAgreement were enlarged  
100 from 35 to 60 characters, CIM WG recommends being careful with the management of  
101 the mRIDs for each specific object.

102 In this particular case, CIM WG recommends using ASCII characters mainly upper case  
103 and lower case letters, digits, underscore (\_) and dash (-). Although XMLs use UTF-8  
104 codification, it can happen that some internal applications or databases are not ready  
105 to manage UTF-8 characters and can have some problems to deal with them.

106

## 107 Usage of revisionNumber

108 There are two compliant ways to make sure that an older document is not replacing a newer  
109 one when providing updates for a document.

110

- 111 • Always a unique mRID, a fixed revision number of 1 and using the createdDateTime
- 112 • Keeping the same mRID for documents, and increasing the revision number

113

114 Which one to use is a business decision.

115

116 The combination of mRID and revision number shall always be unique.

117

118 Always a unique mRID and requiring increasing revision number at the same time is not the  
119 intended use of the revision number and should be avoided!

120

121 A document is considered out-of-order if it has the same or a lower revisionNumber than the  
122 latest received document with the **same mRID**. Likewise, a document with a **new mRID** is  
123 considered out-of-order if it covers a context and timeperiod for which data has previously been  
124 received and it has an older timestamp than the previously received data.

125 Documents received out-of-order should be rejected.

126

127 It is a business decision if documents should replace the previous version completely or only  
128 update certain data.

129

130 As a reply to a received market document an acknowledgement will be submitted. The  
131 acknowledgement document will have a reference back to the acknowledged document. The  
132 information used for the reference is the mRID and revisionNumber of the original document.  
133 The acknowledgement document itself has no revisionNumber and should have a new unique  
134 document mRID.

135

## 136 Usage of the Harmonised Electricity Market Role Model (HEMRM)

137 Please refer to HEMRM documentation. It is recommended to use as much as possible the  
138 HEMRM to describe the data exchanges. In some case (for historical reason or for discussions  
139 outside Europe) the use case of some implementation guides cannot be compliant with the last  
140 version of the HEMRM.

141 It is also obvious that the implementations can refer to the latest version of the HEMRM  
142 available at the implementation time, whereas there are updates and modifications of HEMRM  
143 published after the implementation. Therefore, it is recommended to refer in implementation  
144 guide which HEMRM version the implementation is based on.

145

## 146 Coding scheme identification

147 Whenever a coded value within a MarketDocument is associated with a coding scheme, the  
148 coding scheme shall always be identified. The coding scheme is an independent facet of an  
149 attribute with a size of 3 alphanumeric characters. The coding scheme is provided in Table 1 -  
150 Codelist CodingSchemeType are available in the ENTSO-E codelist  
151 ([https://www.entsoe.eu/publications/electronic-data-interchange-edi-](https://www.entsoe.eu/publications/electronic-data-interchange-edi-library/#Common_information_model__CIM__European_style_market_profile)  
152 [library/#Common\\_information\\_model\\_\\_CIM\\_\\_European\\_style\\_market\\_profile](https://www.entsoe.eu/publications/electronic-data-interchange-edi-library/#Common_information_model__CIM__European_style_market_profile)).

153 **Table 1 - Codelist CodingSchemeType**

| Code | Title                                         | Description                                                                                                                                                                             |
|------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A01  | EIC                                           | The coding scheme is the Energy Identification Coding Scheme (EIC), maintained by ENTSO-E.                                                                                              |
| A02  | CGM                                           | The coding scheme used for Common Grid Model Exchange Standard (CGMES).                                                                                                                 |
| A03  | ISO                                           | The coding scheme for the preceding attribute is a code maintained by International Organization for Standardization (ISO), such as "ISO 3166 English country names and code elements". |
| A10  | Coding scheme which are part of GS1 system    | The coding schemes maintained by GS1.                                                                                                                                                   |
| NAD  | Andorra National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NAL  | Albania National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NAM  | Armenia National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NAT  | Austria National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NAZ  | Azerbaijan National coding scheme             | The National coding scheme of the country in question.                                                                                                                                  |
| NBA  | Bosnia and Herzegovina National coding scheme | The National coding scheme of the country in question.                                                                                                                                  |
| NBE  | Belgium National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NBG  | Bulgaria National coding scheme               | The National coding scheme of the country in question.                                                                                                                                  |
| NCH  | Switzerland National coding scheme            | The National coding scheme of the country in question.                                                                                                                                  |
| NCS  | Serbia and Montenegro National coding scheme  | The National coding scheme of the countries in question.                                                                                                                                |
| NCY  | Cyprus National coding scheme                 | The National coding scheme of the country in question.                                                                                                                                  |
| NCZ  | Czech Republic National coding scheme         | The National coding scheme of the country in question.                                                                                                                                  |
| NDE  | Germany National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NDK  | Denmark National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |
| NEE  | Estonia National coding scheme                | The National coding scheme of the country in question.                                                                                                                                  |



| Code | Title                                 | Description                                                                              |
|------|---------------------------------------|------------------------------------------------------------------------------------------|
| NES  | Spain National coding scheme          | The National coding scheme of the country in question.                                   |
| NFI  | Finland National coding scheme        | The National coding scheme of the country in question.                                   |
| NFR  | France National coding scheme         | The National coding scheme of the country in question.                                   |
| NGB  | United Kingdom National coding scheme | The National coding scheme of the country in question.                                   |
| NGE  | Georgia National coding scheme        | The National coding scheme of the country in question.                                   |
| NGI  | Gibraltar National coding scheme      | The National coding scheme of the country in question.                                   |
| NGR  | Greece National coding scheme         | The National coding scheme of the country in question.                                   |
| NHR  | Croatia National coding scheme        | The National coding scheme of the country in question.                                   |
| NHU  | Hungary National coding scheme        | The National coding scheme of the country in question.                                   |
| NIE  | Ireland National coding scheme        | The National coding scheme of the country in question.                                   |
| NIT  | Italy National coding scheme          | The National coding scheme of the country in question.                                   |
| NKG  | Kyrgyzstan National coding scheme     | The National coding scheme of the country in question.                                   |
| NKZ  | Kazakhstan National coding scheme     | The National coding scheme of the country in question.                                   |
| NLI  | Liechtenstein National coding scheme  | The National coding scheme of the country in question.                                   |
| NLT  | Lithuania National coding scheme      | The National coding scheme of the country in question.                                   |
| NLU  | Luxembourg National coding scheme     | The National coding scheme of the country in question.                                   |
| NLV  | Latvia National coding scheme         | The National coding scheme of the country in question.                                   |
| NMA  | Morocco National coding scheme        | The National coding scheme of the country in question.                                   |
| NMD  | Moldavia National coding scheme       | The National coding scheme of the country in question.                                   |
| NMK  | Macedonia National coding scheme      | The National coding scheme of the country in question.                                   |
| NNL  | Netherlands National coding scheme    | The National coding scheme of the country in question.                                   |
| NNN  | Nordic Regional coding scheme         | The coding scheme of the Nordic region which covers Denmark, Finland, Norway and Sweden. |
| NNO  | Norway National coding scheme         | The National coding scheme of the country in question.                                   |
| NPL  | Poland National coding scheme         | The National coding scheme of the country in question.                                   |
| NPT  | Portugal National coding scheme       | The National coding scheme of the country in question.                                   |

| Code | Title                                     | Description                                            |
|------|-------------------------------------------|--------------------------------------------------------|
| NRO  | Romania National coding scheme            | The National coding scheme of the country in question. |
| NRU  | Russian Federation National coding scheme | The National coding scheme of the country in question. |
| NSE  | Sweden National coding scheme             | The National coding scheme of the country in question. |
| NSI  | Slovenia National coding scheme           | The National coding scheme of the country in question. |
| NSK  | Slovakia National coding scheme           | The National coding scheme of the country in question. |
| NTR  | Turkey National coding scheme             | The National coding scheme of the country in question. |
| NUA  | Ukraine National coding scheme            | The National coding scheme of the country in question. |